

AI Room Reshuffle & Redesign App Flow

A structured product flow for an AI-powered app that converts real room video into reshuffle layouts, redesign concepts, editable measurements, and fit-aware furniture recommendations.

Core Product Concept

An AI-powered room planning app that lets users record or upload a video of their room, detects furniture and room structure, then generates practical reshuffle or redesign options with editable measurements and user preferences.

Important Reality Check

Exact room measurement from arbitrary video is technically risky. The app should position dimensions as estimated, require guided capture, and always allow user correction. This protects user trust and makes the product actually usable.

Mode	Purpose	Complexity	Best for MVP?
Reshuffle	Use existing furniture and generate better layouts.	Medium	Yes
Redesign	Recommend new furniture/decor based on measurements, budget, and style.	High	Later

Document Map

- 1 App entry and mode selection flow
- 2 Flow A: Reshuffle existing room
- 3 Flow B: Redesign room with measurements and products
- 4 Main navigation and saved projects
- 5 Recommended MVP roadmap
- 6 Key screens and product rules
- 7 Positioning and app name ideas

1. App Entry Flow

The first screen should split users into two clear paths. Do not mix reshuffle and redesign too early, because the user intent and backend complexity are different.

User Choice	Meaning	Expected Output
Reshuffle My Room	User already owns the furniture/items and wants better placement.	4-5 layout images, top-down plan, move instructions.
Redesign My Room	User wants new furniture, decor, and product suggestions.	4-5 redesign concepts, measurements, product list, estimated cost.

Flow A: Reshuffle Existing Room

This should be the first MVP flow because it avoids shopping integrations, live inventory, affiliate logic, and exact product dimension matching.

2A. User Selects Reshuffle My Room

The app starts with capture instructions to increase AI accuracy.

- Record slowly and avoid fast camera movement.
- Capture all corners of the room.
- Show the floor, walls, windows, and doors.
- Make sure major furniture is visible.
- Optional: place a known reference object for scale.

3A. Upload or Record Video

Action	Description
Record Video	User records inside the app with guided capture prompts.
Upload Video	User selects an existing video from the gallery.
Processing State	System shows “Analyzing your space...” while detection runs.

4A. AI Detects Room Items

The AI detects visible room objects, then shows them for review. The review step is critical because object detection will not be perfect.

Detected Item	User Controls
Sofa, chair, coffee table, TV stand, bed, desk, lamp, shelf, plant, rug, window, door	Rename item, delete wrong item, add missing item, mark as movable or fixed.

5A. User Sets Reshuffle Preferences

Preference Type	Options
Goal	More open space, better TV setup, better work/study setup, better walking path, better lighting, more storage, better aesthetic.
Fixed Items	TV, bed, desk, wardrobe, wall-mounted items, heavy furniture.
Style	Minimal, cozy, modern, functional, luxury, student apartment, small-space optimized.

6A. AI Generates Layout Options

The app should generate 4-5 layout suggestions. Each result needs more than an image; it should explain what changed and why.

Output Element	Purpose
Generated room image	Shows the reshuffled visual result.
Top-down layout	Makes item placement easier to understand.
Reasoning	Explains why the layout works.
Pros and cons	Helps user compare layouts.
Difficulty level	Sets realistic expectations for moving items.

7A. User Selects or Modifies Layout

Users can save, compare, regenerate, or ask AI to modify a layout.

- Make it more spacious.
- Keep the sofa where it is.
- Move the desk near the window.
- Make it look more modern.
- Use only existing furniture.

8A. Final Reshuffle Plan

The final output should include a before/after comparison and step-by-step instructions, not just pretty images.

- 1 Move coffee table closer to the sofa.
- 2 Rotate desk toward the window.
- 3 Place floor lamp beside the sofa.
- 4 Shift rug under the seating area.
- 5 Keep TV stand unchanged if marked fixed.

Flow B: Redesign Room

This is the advanced flow. It requires room measurement estimation, user correction, product recommendations, budget filtering, and dimension matching.

2B. User Selects Redesign My Room

Question	Example Options
What do you want to redesign?	Bedroom, living room, studio apartment, office, kitchen area, dining area, full apartment section.
What should AI improve?	Full redesign, add furniture, improve decor, lighting, premium look, budget-friendly redesign, small-space optimization.

3B. Upload or Record Video

Redesign mode needs stricter capture guidance because measurements and product fit depend on visual accuracy.

- Capture all walls and corners.
- Show floor and ceiling height where possible.
- Capture windows, doors, outlets, and current furniture.
- Slowly scan from each corner for better geometry estimation.

4B. AI Detects Room Structure

The system identifies structural features and usable space before recommending design changes.

- Room boundaries and wall surfaces.
- Floor area and open zones.
- Windows and doors.
- Current furniture and empty spaces.
- Potential usable wall space.

5B. Measurement Detection and Review

Measurements must be displayed as estimates. The user must be able to edit every important dimension.

Area	AI Estimate	User Action
Room Width	11.5 ft	Edit
Room Length	14 ft	Edit
Wall Height	8 ft	Edit
Window Width	4 ft	Edit
Door Width	3 ft	Edit

6B. Optional Calibration Step

To improve measurement accuracy, ask the user to confirm one known object measurement. This could be a door, A4 paper, credit card, bed, table, or any manually measured object.

Example: “What is the width of this door?” User enters 36 inches, then the AI recalibrates the room estimates.

7B. User Sets Redesign Preferences

Preference	Options
Budget	Under \$250, \$250-\$500, \$500-\$1,000, \$1,000-\$2,500, custom.
Style	Modern, minimal, cozy, luxury, Scandinavian, industrial, boho, student apartment, Indian home style, gaming setup, work-from-home setup.
Color	Neutral, warm, dark, light, earth tones, custom.
Stores	Amazon, IKEA, Walmart, Wayfair, Target, Home Depot, local stores, no preference.
Must-have Items	Desk, sofa, bed, storage, bookshelf, TV unit, dining table, rug, lighting, plants, wall art.

8B. AI Generates Redesign Concepts

The AI generates 4-5 design concepts. Each concept should combine visuals, budget, fit confidence, and product reasoning.

Concept Element	Description
Rendered design image	Shows the redesigned room.
Top-down layout	Shows where items go.
Product list	Recommended furniture and decor.
Estimated total cost	Based on selected budget and products.
Fit confidence	How likely the products fit the measured room.
Alternatives	Cheaper or style-matched replacement items.

9B. Product Recommendation Flow

Product cards should show dimensions clearly. A good-looking product is useless if it does not fit.

Product	Size	Price	Fit
Compact Sofa	72 in wide	\$399	Fits
Coffee Table	40 x 22 in	\$89	Fits
Rug	6 x 9 ft	\$129	Fits

10B. User Customizes Design

- Replace product.
- Change color or style.
- Lower budget.
- Remove or add item.
- Move item or lock item position.
- Regenerate the design.

Useful prompts: “Make this under \$500,” “Use a smaller sofa,” “Add a study desk,” “Keep my current bed,” or “Show IKEA-only products.”

11B. Final Redesign Plan

The final redesign should include the final image, top-down layout, shopping list, total cost, setup instructions, measurement notes, and saved project state.

Main App Navigation

Section	What It Contains
Home	Start new project, recent room scans, saved designs, reshuffle history.
Projects	Room video, detected items, measurements, generated designs, saved layouts, shopping list.
Explore Styles	Minimal bedroom, cozy living room, small apartment, work-from-home setup, budget redesign, luxury redesign.
Saved Designs	Saved layouts, product lists, favorite designs, before/after images.
Profile / Preferences	Budget preference, favorite stores, style preference, measurement units.

Recommended MVP Roadmap

Do not build the full redesign system first. Start with reshuffle because it delivers value without product catalog complexity.

Version	Scope	Features
MVP V1	Reshuffle Existing Room	Upload video/images, detect items, confirm items, select preferences, generate 3-5 reshuffle images, save design.
V2	Measurement + Redesign	Room measurement detection, correction screen, redesign mode, furniture recommendations, budget filter, product dimension matching.
V3	Advanced Features	AR preview, drag-and-drop editor, real-time item replacement, multi-room support, designer marketplace, affiliate links, premium designs.

Key Screens

#	Screen
1	Welcome Screen
2	Mode Selection Screen
3	Capture Instructions Screen
4	Upload / Record Screen
5	Video Processing Screen
6	Detected Items Review Screen
7	Measurement Review Screen
8	Preference Selection Screen
9	AI Design Results Screen
10	Design Detail Screen
11	Product Recommendation Screen
12	Product Swap Screen

#	Screen
13	Final Plan Screen
14	Saved Projects Screen

Important Product Rules

Rule	Reason
Always let users correct AI output.	AI will make mistakes with objects, dimensions, and room boundaries.
Never claim exact measurements from video.	Use “estimated measurement” to avoid false trust.
Start with images before full 3D.	3D editing is heavier and slower to build.
Reshuffle is easier than redesign.	Redesign requires product search, live prices, dimensions, and store integrations.
Fit accuracy is the trust layer.	One bad furniture recommendation can ruin user confidence.

Full User Flow Summary

Reshuffle Flow

- 1 User opens app.
- 2 Chooses Reshuffle My Room.
- 3 Uploads or records video.
- 4 AI detects room items.
- 5 User confirms or edits detected items.
- 6 User selects goals and preferences.
- 7 AI generates 4-5 layout images.
- 8 User compares options.
- 9 User asks for changes if needed.
- 10 User saves final layout.
- 11 App provides step-by-step reshuffle instructions.

Redesign Flow

- 1 User opens app.
- 2 Chooses Redesign My Room.
- 3 Uploads or records video.
- 4 AI detects room structure.
- 5 AI estimates room measurements.
- 6 User reviews and corrects measurements.
- 7 User selects budget, style, and stores.

- 8 AI generates 4-5 redesign concepts.
- 9 User views recommended products.
- 10 User swaps, removes, or changes products.
- 11 AI regenerates design.
- 12 User finalizes design.
- 13 App provides shopping list and setup plan.

Positioning and Naming

Do not position this only as “AI interior design.” That category is crowded. The stronger positioning is video-based reshuffling plus fit-aware redesign.

Recommended positioning:

AI room reshuffling and redesign from real video, with editable measurements and fit-aware furniture suggestions.

One-line product concept:

An AI-powered room planning app that turns a room video into reshuffle layouts, redesign concepts, measurements, and furniture recommendations that actually fit the space.

Name Ideas

Name	Fit
ReSpace AI	Clean and clear. Best overall balance.
RoomShift	Strong fit for reshuffling.
DwellAI	More premium redesign feeling.
Layoutly	Simple and product-focused.
HomeShuffle	Very literal, but less premium.

Recommended Build Order

Priority	Build This First	Why
1	Video upload + room/item detection review	This validates whether users can get useful room understanding.
2	Preference intake + reshuffle generation	This gives users quick visible value.
3	Saved projects + final move checklist	This makes the app actionable, not just visual.
4	Measurement correction	Needed before serious redesign/product matching.
5	Product recommendations	Only worth building after measurement flow is reliable.

Bottom line: Start with AI reshuffling. Prove that users can upload a room video, confirm detected items, and get useful layout options. Once that works, add measurement correction and product-based redesign.